

Theorem 1. *There are infinitely many prime numbers.*

Proof. Let $\mathbb{P}$ be the set of prime numbers. Suppose to the contrary that $\mathbb{P}$ is finite. Then we write

$$\mathbb{P} = \{p_1, p_2, \dots, p_n\}.$$

Set $q = p_1 p_2 \cdots p_n + 1$. Then q is not divisible by any prime number in $\mathbb{P}$. Thus, every prime factor of q is not in $\mathbb{P}$, a contradiction.

□